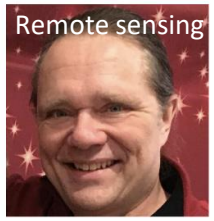
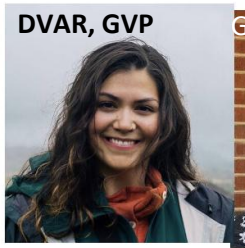
About Me and Jo

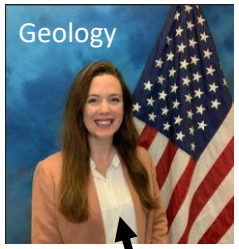


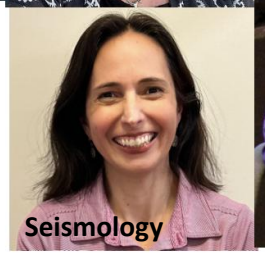






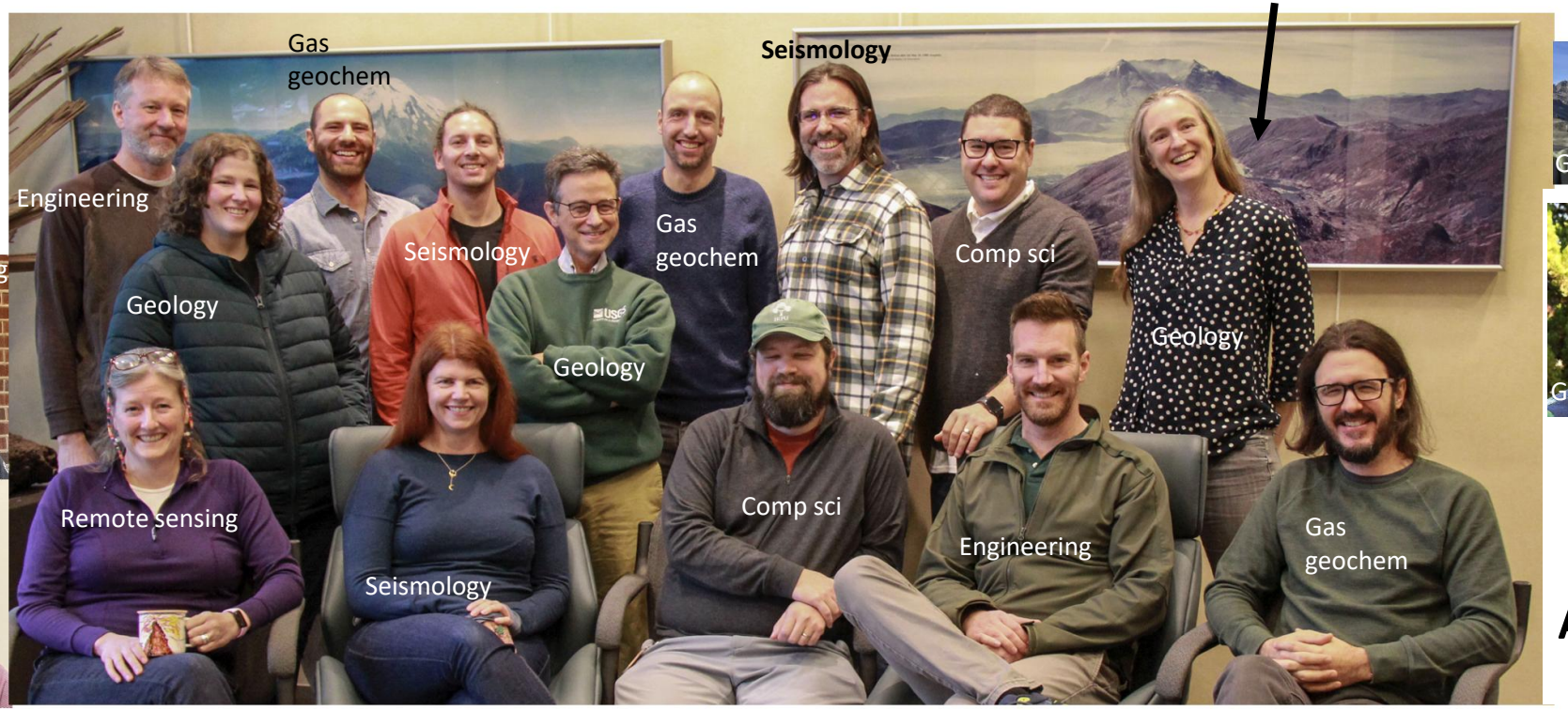






State Dept. Liaison
Cassandra Smith

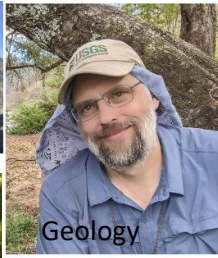
Director
Heather Wright



Associate director
Angie Diefenbach









Associate director
Dave Ramsey

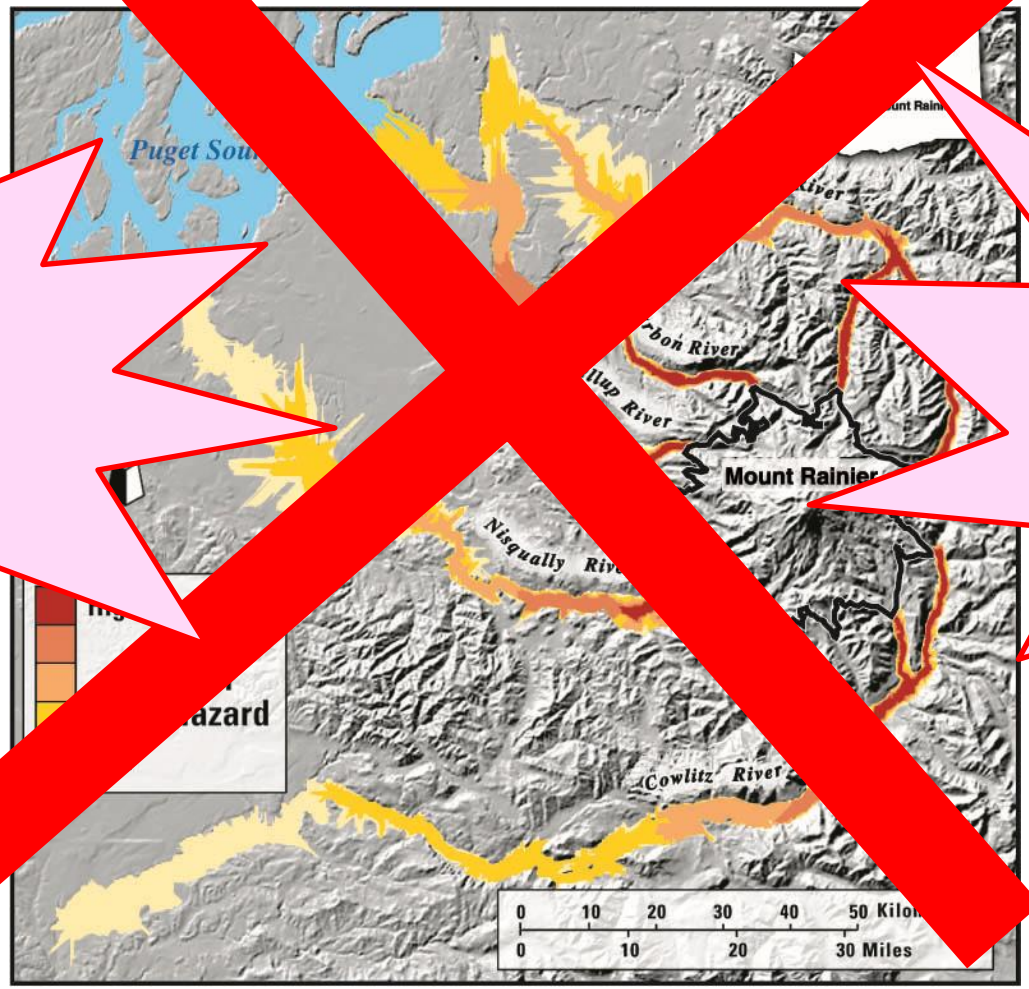
VDAP
Volcano Disaster Assistance Program

USGS
science for a changing world

VDAP is an international volcano crisis response team funded by US Geological Survey and the State Department. VDAP is based out of the USGS Cascades Volcano Observatory in Vancouver, Washington.

About the course

CSAV ~~LAHARZ~~



Expanded course!

New this year:
more models
more hazards!!!

VDAP

USGS

About the course

CSAV Hazard Modeling and Map Making

- Hazard map making and design
- Lahar, PDC, & ballistic processes and modeling
- GIS
- VICTOR platform
- Class project

Please be patient; there might be bugs 🥺



VDAP

USGS

About you

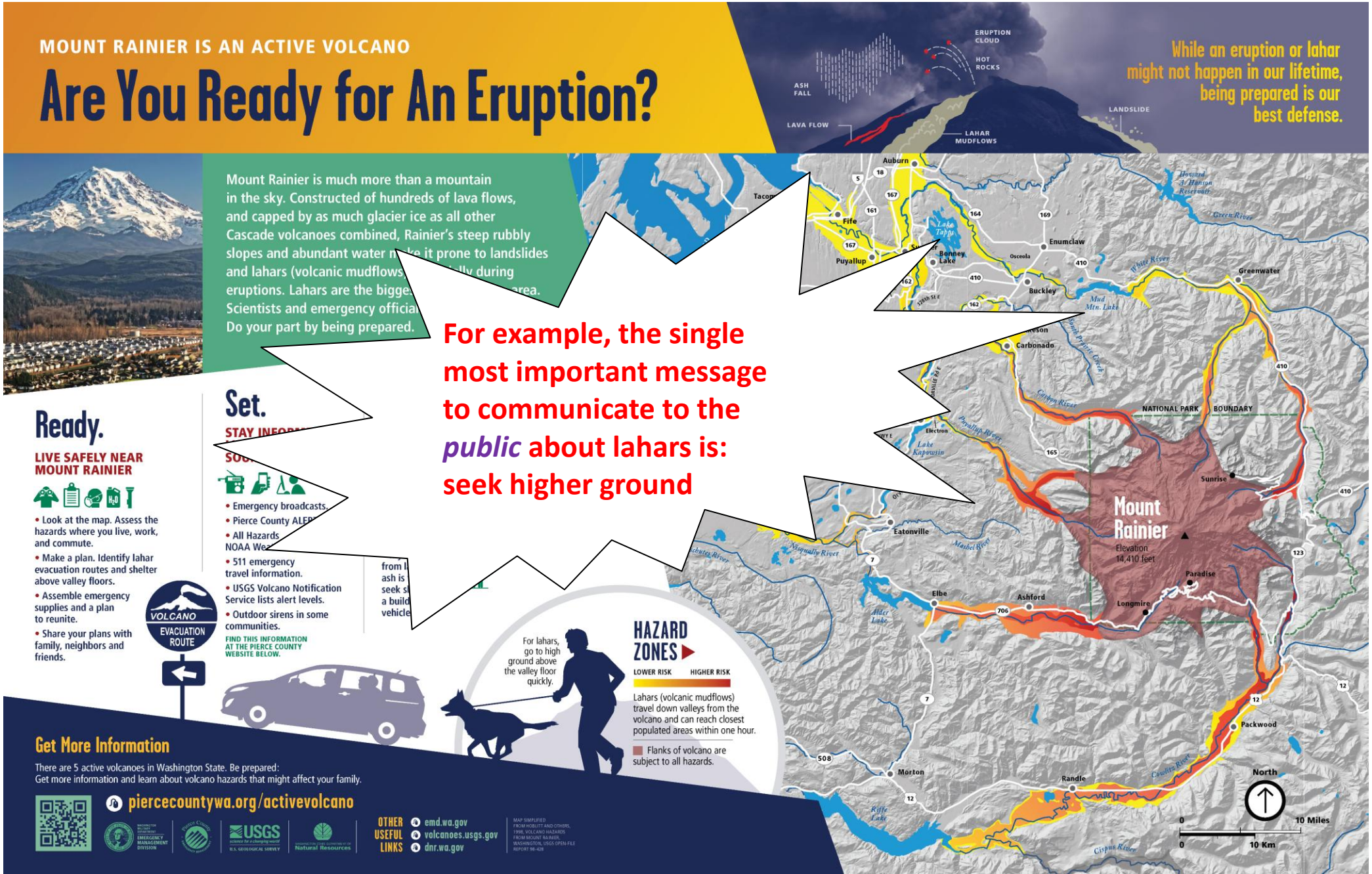
Let’s do some introductions:

- your name
- what country you are from
- what agency you work for
- what your main job is (field technician, seismologist, geologist, etc.)

About you

- ArcMap? ArcGIS Pro? QGIS?
- Lahar modeling? What models?
- Pyroclastic flow modeling? What models?
- Ballistics modeling? What models?
- Hazard-map making?
- VICTOR platform?

End goal of this class is to use models to make a map. But really, the end goal is communication to the map user.



For example, the single most important message to communicate to the *public* about lahars is: seek higher ground



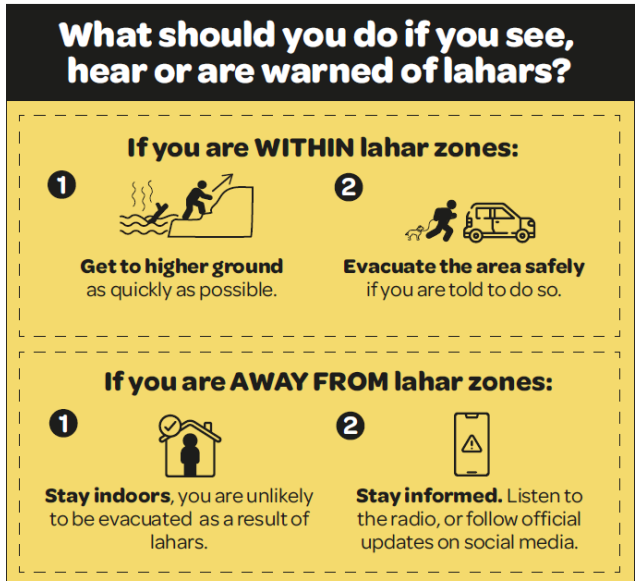
Washington Department of Natural Resources. (2014). Mount Rainier is An Active Volcano, Are You Ready for An Eruption



Volcano Disaster Management Councils of Iwatesan. (2019). Volcanic disaster prevention map of Iwatesan volcano.



Inawashiro Town. (2016). Adatarayama Volcano Disaster Prevention Map.



Taranaki Emergency Management. (2022). Taranaki Maunga. Lahars from Taranaki Version 1.0.

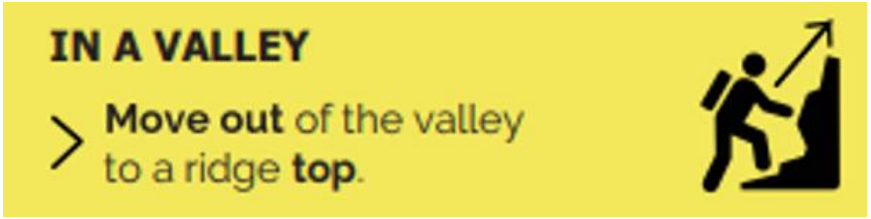


Si te encuentras en el volcán, evita caminar por las barrancas

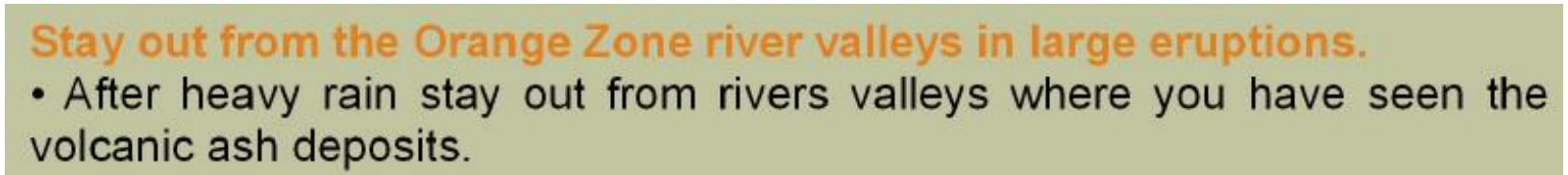


Resguárdate en lugares altos

IG-UNAM, CENAPRED, SEGOB. (2016). Popocatépetl - Lahares.

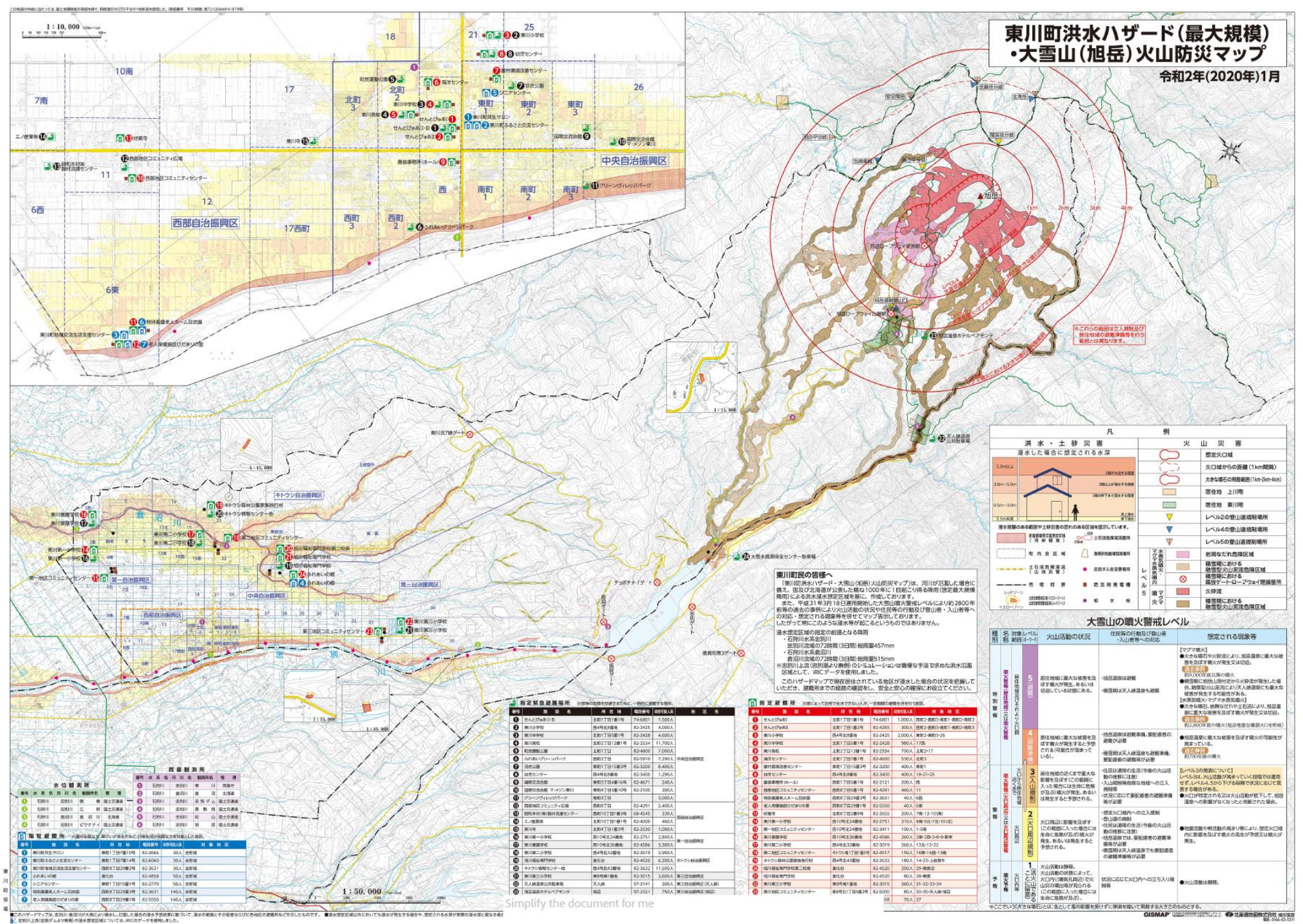


GNS Science (compiler). (2019). Volcanic Hazards at Whakapapa Mt. Ruapehu. Version 6.0 2019.



Vanuatu Meteorology & Geo-Hazards Department. (2019). Ambae Long-Term Background Volcanic Hazards Map.

This community map has much more detailed information and is designed for emergency management and public safety during a crisis



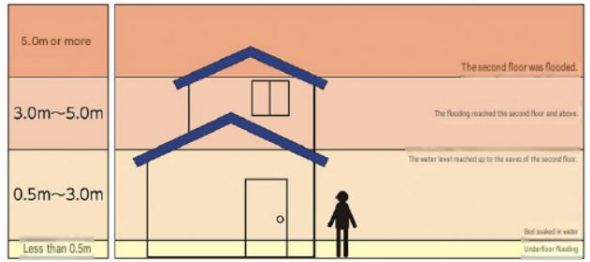

町民運動公園 5
北町 2
北町 3
17
6 海洋センター
7 農村環境改善センター
7 羽衣公園
5 シニアセンター
東町 1
東町 2
東川中学校 3 4
東川高校 4 5
せんとぴゅあI 1
せんとぴゅあ(I・II) 1
せんとぴゅあII 2
1 東川町共生サロン
2 東川町ふるさと交流センター
東川寺 15
国際交流
農協事務所(ホール) 9
西
南町 1
南町 2
南町 3
西町 3
西町 2
6 ふれあいグリーンパーク
17西町
振興区

Fan

example

Flood/landslide disaster

Expected water depth in the event of flooding



This map shows areas that have experienced flooding and areas at risk of landslides.

- Areas where house collapse and other damage are anticipated due to flooding (riverbank erosion)
- Debris flow-prone stream areas (basin, flood domain)
- Neighborhood association area
- Areas at risk of landslides on steep slopes
- flow-prone streams (mountain disasters)
- Chubetsu Dam Discharge Warning Station
- Municipality world
- Disaster prevention generator
- red zone / yellow zone — Landslide hazard zone (yellow zone) Landslide hazard special hazard zone (red zone)
- water hydrant

volcanic disaster

- Estimated crater area
- Distance from the crater area (at 1km intervals)
- Area of large volcanic rock dispersion (1km, 2km, 4km)
- Place of residence: Kamikawa Town
- Place of residence: Higashikawa Town
- Level 2 hiking trail restrictions
- Level 4 hiking trail restrictions
- Level 5 hiking trail restrictions

level 5

Phreatic eruption/ phreatomagmatic eruption

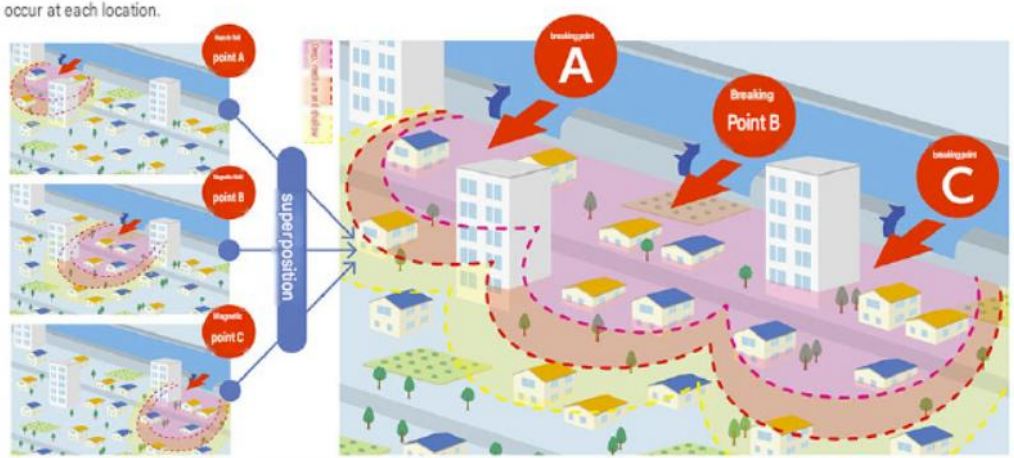
- Debris avalanche risk area
- Areas at risk of snowmelt-induced volcanic mudflows during the snowy season
- Existing gates and ropeway closures during the snowy season

magma eruption

- pyroclastic flow
- Areas at risk of snowmelt-induced volcanic mudflows during the snowy season

《Simulation method》

For every potential breach point (where the levee is expected to break), we conduct flood analysis simulations to predict how flooding and inundation will occur at each location.



The map shows the maximum area and depth when all predicted results are superimposed. In reality, only a limited number of these areas will flood, and the flooded area will expand.

《How floods occur》

Floods occur when heavy rain causes rivers to swell, leading to the collapse of levees or the river water overflowing the levees.



When heavy rain causes the river water level to rise to the top of the embankment, water pressure begins to build up on the embankment.

As the water level rises, the levee can no longer withstand the force of the water, and parts of the levee begin to collapse.

The area where the levee collapsed widens rapidly, and water rushes out with great force, plunging into homes.

Designated emergency evacuation site

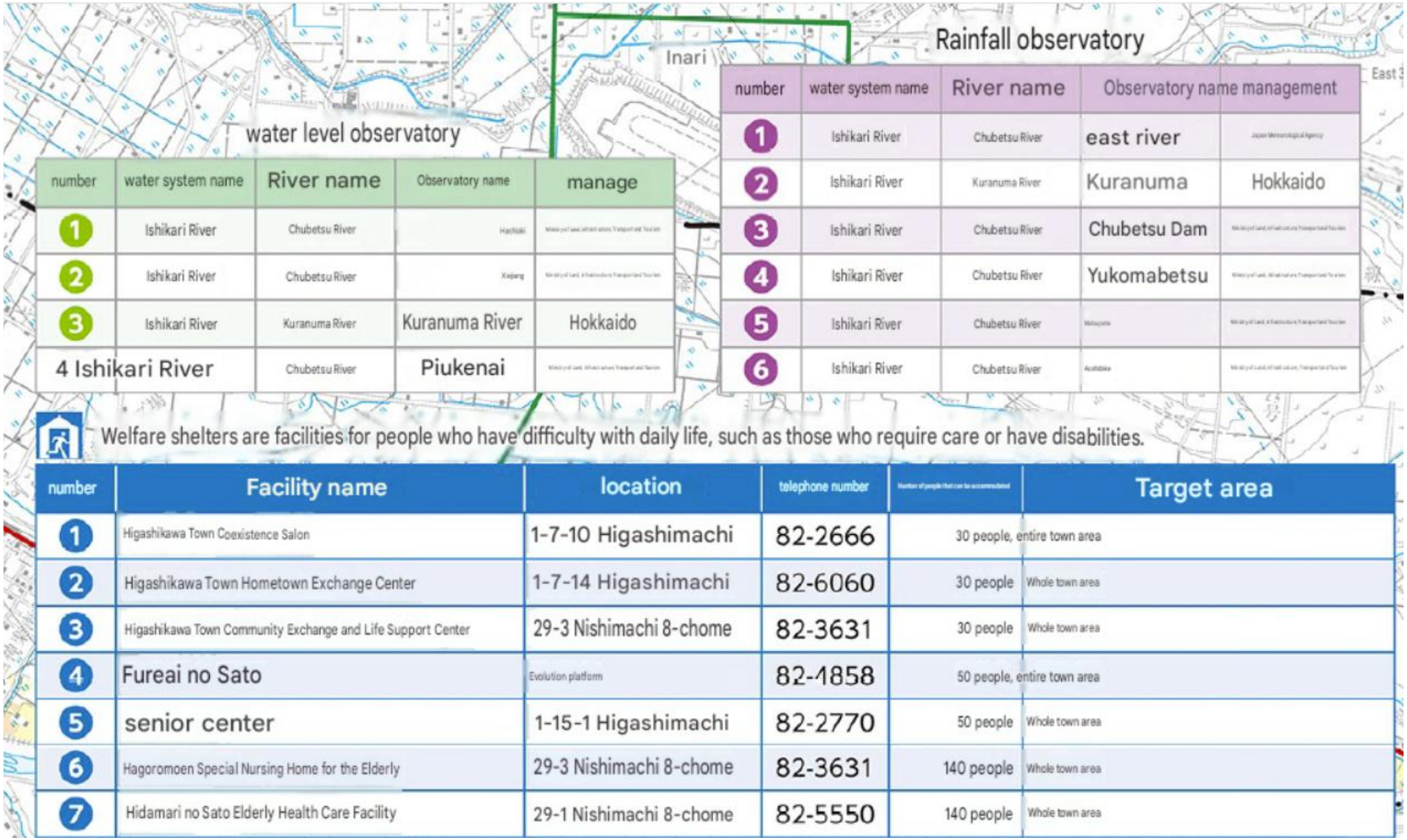
A place to take temporary refuge to avoid danger during a disaster.

number	Facility name	location	telephone number		area name
1. Sentopia (I & II)		1-1-1 Kitamachi	74-6801	1,500 people	
② Higashikawa Elementary School		Nishi 4-go Kita 8-banchi	82-2425	4,000 people	
❸	Higashikawa Junior High School	1-5-1 Kitamachi	82-2428	4,600 people	
④ Higashikawa High School		2-12-1 Kitamachi	82-2534	11,700 people	
⑤ Townspeople's Sports Park		Kitamachi 1-chome	82-4600	7,000 people	
6 Fureai Green Park		Nishimachi 2-chome	82-5910	7,290 people	Central Autonomous Revitalization Region
❼	Hagoromo Park	1-15-3 Higashimachi	82-3200	8,400 people	
③ Early childhood center		Nishi 4-go Kita 8-banchi	82-3400	1,290 people	
❾	International Exchange Center	3-4-16 Higashimachi	82-4671	240 people	
① International Exchange Hall Ma Maison Higashikawa		4-3-10 Higashimachi	82-2100	200 people	
① Green Village Park		Minamimachi 3-chome		3,000 people	
⓬	Western District Community Square	Nishimachi 8-chome	82-4291	2,400 people	
⓭	Showa Lumber Co., Ltd. Sawmill Distribution Center	1-3, Nishimachi 10-chome	68-4545	330 people	Western Autonomous Revitalization Region
① Eno Sangyo Co., Ltd.		1-1-1 Kitamachi 10-chome	82-4000	460 people	
① Tosenji Temple		4-1-3 Kitamachi	82-2530	1,000 people	
⓰	Higashikawa Daiichi Elementary School	24th North, West 10th Street	82-2751	2,800 people	
① Dongchuan Nursing School		Nishi 10-go Kita 36-banchi	82-4586	3,300 people	First Autonomous Revitalization Region
⓲	Higashikawa Daini Elementary School	West 4th Street, North 32	82-3019	2,900 people	
⓳	Asahikawa Welfare College	Evacuation platform	82-4520	6,200 people	Kitōshi Self-Governing Development District
20 Kitōshi Information Center and others		West 4th Street, North 43	82-2632	11,500 people	
㉑	Higashikawa Dokan Elementary School	East 8th Street, South 1st Address	82-3015	2,600 people	Third Autonomous Revitalization Region
㉒	Tenninkyo Onsen Public Parking Lot	Tunnel Gorge	97-2141	200 people	The Third Autonomous Revitalization Zone (Tenninkyo)
㉓	Asahidake Onsen Hotel Bearmonte	Asahidake	97-2321	750 people	Third Self-Governing Development District (Asahidake)
② Heavy Snow Water Resource Conservation Center Parking Lot		Nokanan		200 people	The Third Autonomous Development Zone (ノカナン)

designated shelter

A facility where people who cannot live in their homes due to a disaster can stay for a certain period of time.

number	Facility name	location	telephone number		Target area
1 Sentopia I		1-1-1 Kitamachi	74-6801	1,000 people	Nishimachi 2-3, Minaminachi 1, Minaminachi 2, Minaminachi 3
② Sentopia II		1-1-2 Kitamachi	82-4265	300 people	Nishimachi 2, Nishimachi 3, Minaminachi 1, Minaminachi 2, Minaminachi 3
❸	Higashikawa Elementary School	Nishi 4-go Kita 8-banchi	82-2425	2,000 people	Higashimachi 2, Higashimachi 3-26
❹	Higashikawa Junior High School	1-5-1 Kitamachi	82-2428	980 people	17 West
③ Higashikawa High School		2-12-1 Kitamachi	82-2534	700 people	Kitamachi 2-17
6 Marine Center		1-7-1 Kitamachi	82-4600	500 people	Kitamachi 3
❼	Rural Environment Improvement Center	1-15-3 Higashimachi	82-3200	400 people	Higashimachi 1
③ Early childhood center		Nishi 4-go Kita 8-banchi	82-3400	400 people	18・21・25
❾	Agricultural Cooperative Office (Hall)	1-5-1 Nishimachi	82-2121	200 people	West
① Western District Community Center		8-5-1 Nishimachi	82-4291	460 people	11
⓫	Hagoromoen Special Nursing Home for the Elderly	29-3 Nishimachi 8-chome	82-3631	40 people,	6 West
⓬	Hidamari no Sato Elderly Health Care Facility	29-1 Nishimachi 8-chome	82-5550	40 people,	6 bundles
⓭	Kouji Temple	8-2-9 Kitamachi	82-2655	200 people	7 South, 12, 10 (South)
⓮	Higashikawa Daiichi Elementary School	Nishi 10-go Kita 24-banchi	82-2751	310 people	4 South, 5 North, 7 North, 10 (North)
① District 1 Community Center		Nishi 10-go Kita 24-banchi	82-3411	100 people	1.5 South
⓰	Dongchuan nursing school	Nishi 10-go Kita 36-banchi	82-4586	260 people	2 East, 2 West, 3, 8, 9, Shin-ei
① Higashikawa Daini Elementary School		Nishi 4-go Kita 32-banchi	82-3019	260 people	13 North 15-22
⓲	Second District Community Center	1-1-5 Kitōshi Minami	82-4517	150 people	16 East, 16 West, 13 South
⓳	Kitoushi Forest Park Family Travel Village	Nishi 4-go Kita 43	82-2632	180 people	14-23・Kamiki Touyu
⓴	Asahikawa Welfare Vocational School, Second Campus	Evacuation platform ※	82-4520	250 people	29 Higashi Iburi Marsh
㉑	Asahikawa Welfare College	Evacuation platform ※	82-4520	80 people	28・Shinonome
㉒	Higashikawa Dokan Elementary School	East 8th Street, South 1st Address	82-3015	260 people	31・32・33・34
㉓	Third District Community Center	1-5-2 Higashi 8-go Kita	82-5000	80 people,	30-35, Tenninkyo, Asahidake
24 Fureai no Sato		Evacuation platform	82-4858	70 people	27



water level observatory

number	water system name	River name	Observatory name	manage
1	Ishikari River	Chubetsu River	Hachioki	
2	Ishikari River	Chubetsu River	Xiajiang	
3	Ishikari River	Kuranuma River	Kuranuma River	Hokkaido
4 Ishikari River		Chubetsu River	Piukenai	

Rainfall observatory

number	water system name	River name	Observatory name management	
1	Ishikari River	Chubetsu River	east river	Japan Meteorological Agency
2	Ishikari River	Kuranuma River	Kuranuma	Hokkaido
3	Ishikari River	Chubetsu River	Chubetsu Dam	
4	Ishikari River	Chubetsu River	Yukomabetsu	
5	Ishikari River	Chubetsu River		
6	Ishikari River	Chubetsu River		

Welfare shelters are facilities for people who have difficulty with daily life, such as those who require care or have disabilities.

number	Facility name	location	telephone number	Number of people that can be accommodated	Target area
1	Higashikawa Town Coexistence Salon	1-7-10 Higashimachi	82-2666	30 people,	entire town area
2	Higashikawa Town Hometown Exchange Center	1-7-14 Higashimachi	82-6060	30 people	Whole town area
3	Higashikawa Town Community Exchange and Life Support Center	29-3 Nishimachi 8-chome	82-3631	30 people	Whole town area
4	Fureai no Sato	Evolution platform	82-1858	50 people,	entire town area
5	senior center	1-15-1 Higashimachi	82-2770	50 people	Whole town area
6	Hagoromoen Special Nursing Home for the Elderly	29-3 Nishimachi 8-chome	82-3631	140 people	Whole town area
7	Hidamari no Sato Elderly Health Care Facility	29-1 Nishimachi 8-chome	82-5550	140 people	Whole town area











Hazard Map World Cup

Homework- Day One

1. With your team, pick a volcano where you want to make a hazard map.
2. Review the eruption history of your volcano using the Smithsonian Institution GVP website (http://www.volcano.si.edu/) and/or observatory reports.
   - What is typical behavior for an eruption in recent history (VEI, domes, lava flows, lahars, magma type, effusive/explosive)?
   - What are the potential larger, longer-term styles of eruption?
3. Gather information about hazards at your volcano:
   - Lahar, PDC, and ballistics history; volumes; and recurrence frequency (how big? how often?)
   - What are the triggers for lahars (typhoon, rainstorm, earthquake, eruption, landslide, pyroclastic flow, rapid snow melt)?
   - What types of PDCs does the volcano produce? Dome-collapse, column-collapse, lava flow-front collapse, lateral blasts?
   - Where have lahars, PDCs, and ballistics been mapped or reported?
4. Is there already a hazard map?
   - Is there perhaps a good reason to make a new map for a **different scenario, different purpose, or different audience**?
   - Use https://volcanichazardmaps.org/ to look at maps for your volcano and other example maps
5. Discuss what scenario(s) you would like to portray.

Homework- Day Two

1. Review any spatial information about your volcano.
    - Geologic map: where are the pyroclastic flows, lava flows, lahars, etc.
    - Use Google Earth to explore your volcano
    - Where are the steep areas? Are the channels broad, shallow, narrow, deep? What is the summit like? Pointy and conical? Is there a deep crater? Is it open to one side?
    - Make an elevation profile and think about the H/L value you might use for LAHARZ
2. Where are the people?
    - Find a population map. How many people live nearby?
3. Identify major roads, rivers, lakes, villages, major cities. Where are the most important, critical places if a disaster happens?
    - Airports, energy resources, military assets, hospitals, schools and evacuation centers, sensitive natural areas
4. Decide on the scenario or scenarios for hazard map:
    - Are you considering worst-case and/or most-likely scenarios? Or a range?
    - What type of lahars are you portraying (rain-fall triggered, eruption triggered, etc?)
    - What types of PDCs are you portraying (dome-collapse, column-collapse, etc?)
    - What size/volume eruption(s), lahars, and PDCs are you considering?
5. Decide on your target audience:
    - General public? Mining company or transportation agency? Tourists? Emergency managers? Land-use planners?

Things to Consider

- Map should be beautiful and easy to understand.
- Put your names on it. Stake your professional reputation on it.
- Logos for partner agencies; make it authoritative.
- Scale bar, north arrow, location map or spatial context
- Metadata, cite references or make acknowledgements to other data sources
- What is your message?
- Who is your audience? Technical or non-technical?
- Are you making a map focused on hazard level (high-moderate-low, hazards are combined) or focused on the type of hazard (PDC, lahar, etc.)?
- Appropriate colors with good design elements
- Areas of relative hazard easy to understand (high, medium, low) or different hazard processes easy to distinguish
- What is your level of uncertainty?
- Oblique photo and/or 3-D view are optional but helpful
- Be creative and have fun!